

Database

Introduction to the Concepts of Database Management System (DBMS)

A **Database Management System (DBMS)** is software that helps create, manage, organize, and retrieve data efficiently in a structured way.

Key concepts:

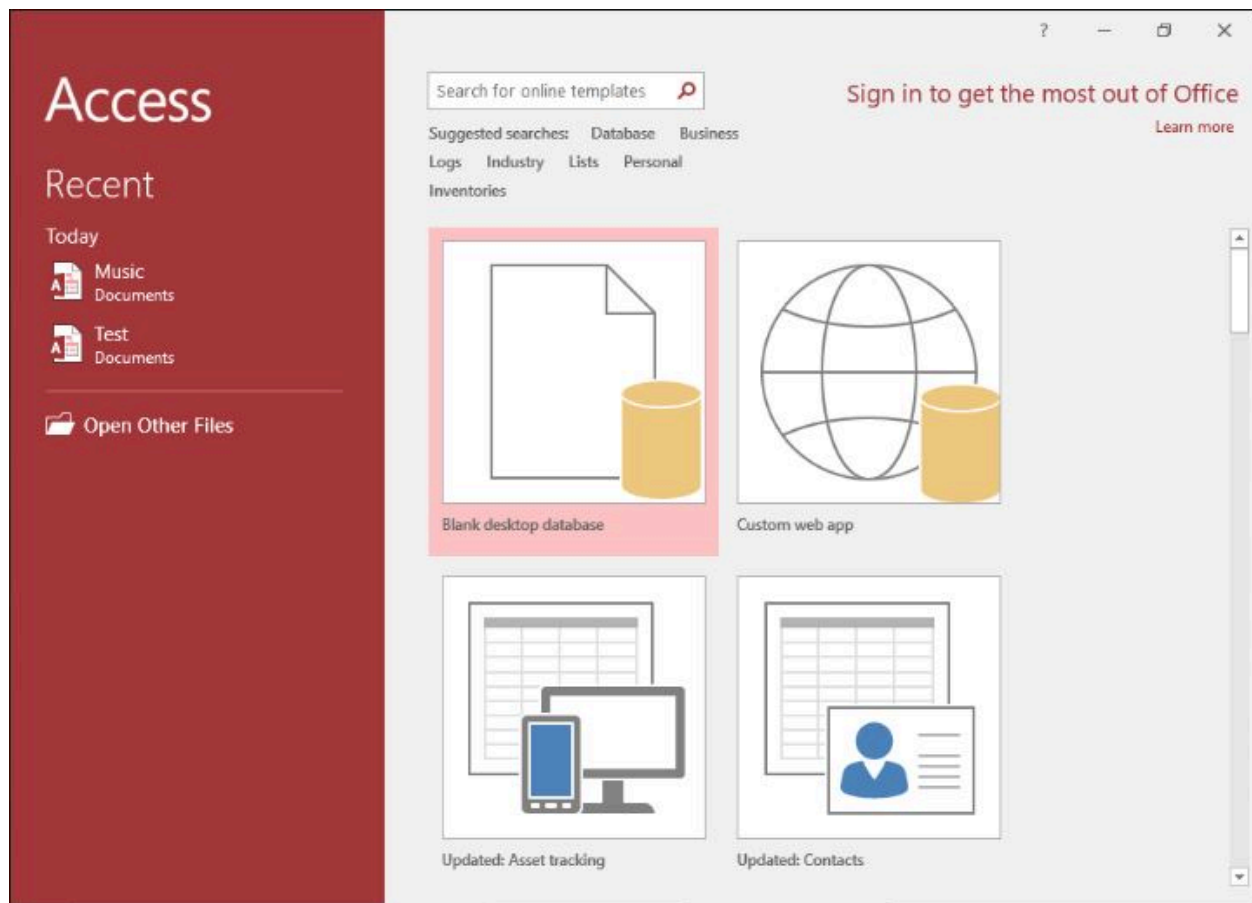
- **Database** → A collection of related data stored in an organized manner (e.g., student records, employee details).
- **Table** → Basic unit; like a spreadsheet with rows (records) and columns (fields).
- **Field** → A single piece of information (column), e.g., Name, Roll No., Date of Birth.
- **Record** → A complete row of data about one entity (e.g., one student's full details).
- **Primary Key** → Unique identifier for each record (e.g., Roll No. or Student ID).
- **Query** → Question asked to the database to retrieve, filter, or calculate data.
- **Form** → User-friendly interface for entering/viewing data (like a data entry screen).
- **Report** → Formatted output for printing or viewing summarized data.

Advantages of DBMS: Data security, reduced redundancy, easy backup, multi-user access, and fast searching.

Here are examples of the Microsoft Access main interface when you first open it (showing blank database creation options):



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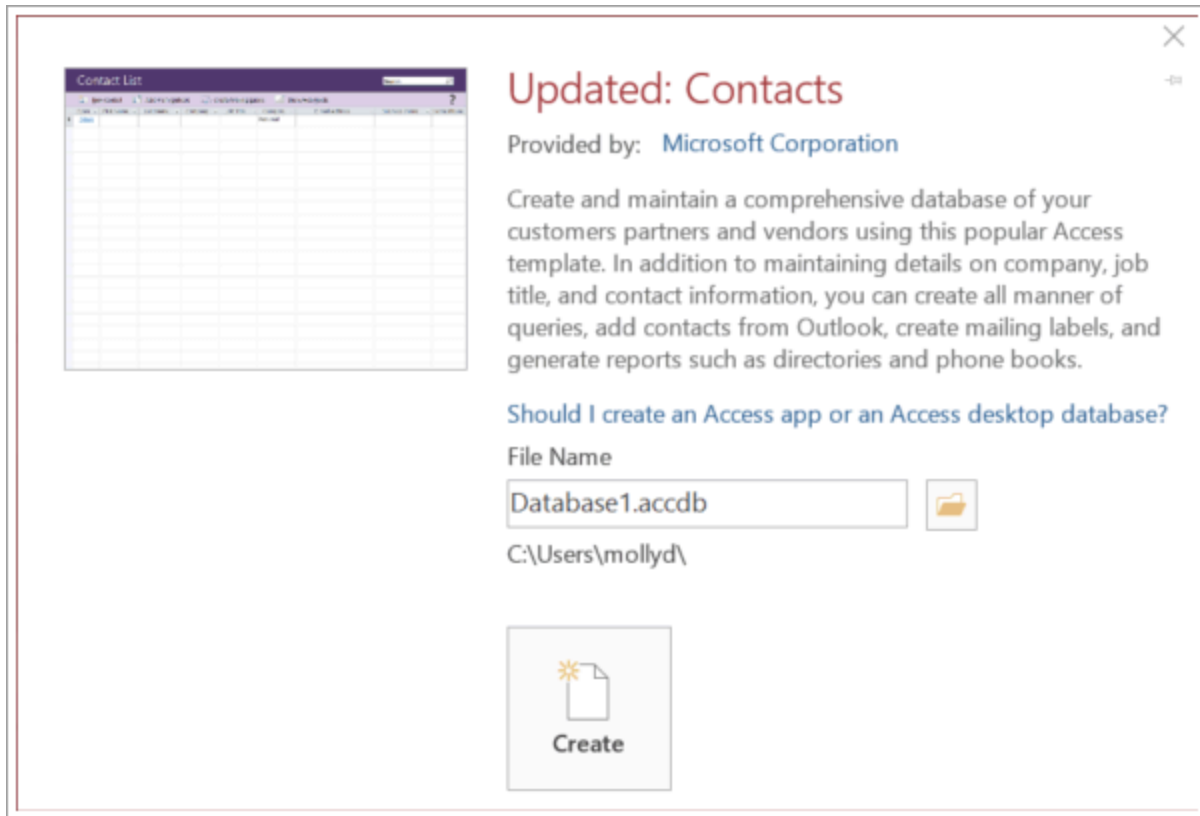


mlpp.pressbooks.pub

Creating a Database

1. Open **Microsoft Access**.
2. On the start screen → **Blank desktop database** (or choose a template).
3. Enter a **File Name** (e.g., SchoolDatabase.accdb).
4. Choose location → Click **Create**.

Here is a visual example of creating a new blank database:



support.microsoft.com

Video: Create an Access desktop database - Microsoft Support

Creating a Table: Concepts of Field, Field Types

Tables are created in two main ways: **Datasheet View** (easy, like Excel) or **Design View** (recommended for control).

- **Field** → Column name (e.g., StudentID, Name).
- **Common Field Types (Data Types):**
 - Short Text → Names, addresses (up to 255 characters).
 - Long Text → Large descriptions.
 - Number → For calculations (Integer, Double, etc.).
 - Date/Time → Dates and times.
 - Yes/No → Checkbox (True/False).
 - AutoNumber → Automatic unique number (great for Primary Key).

Here are examples of creating a table in **Design View** (showing fields, data types, and primary key setting):

The screenshot displays the Microsoft Access interface in Design View for a table named 'Albums'. The ribbon at the top includes 'File', 'Home', 'Create', 'External Data', 'Database Tools', and 'Table Tools'. The 'Table Tools' ribbon has tabs for 'Design' and 'Table Design Tools'. The 'Design' tab is active, showing options for 'Filter', 'Sort & Filter', 'Records', 'Find', and 'Text Formatting'. The 'Field Name' column lists the fields: AlbumId, AlbumName, ReleaseDate, ArtistId, and GenreId. The 'Data Type' column shows the corresponding data types: AutoNumber, Short Text, Date/Time, Number, and Number. The 'Description (Optional)' column is empty. The 'Field Properties' pane is open for the 'AlbumId' field, showing the 'Required' property set to 'Yes'. An orange arrow points to the 'Required' dropdown. The status bar at the bottom indicates 'Design view. F6 = Switch panes. F1 = Help.'

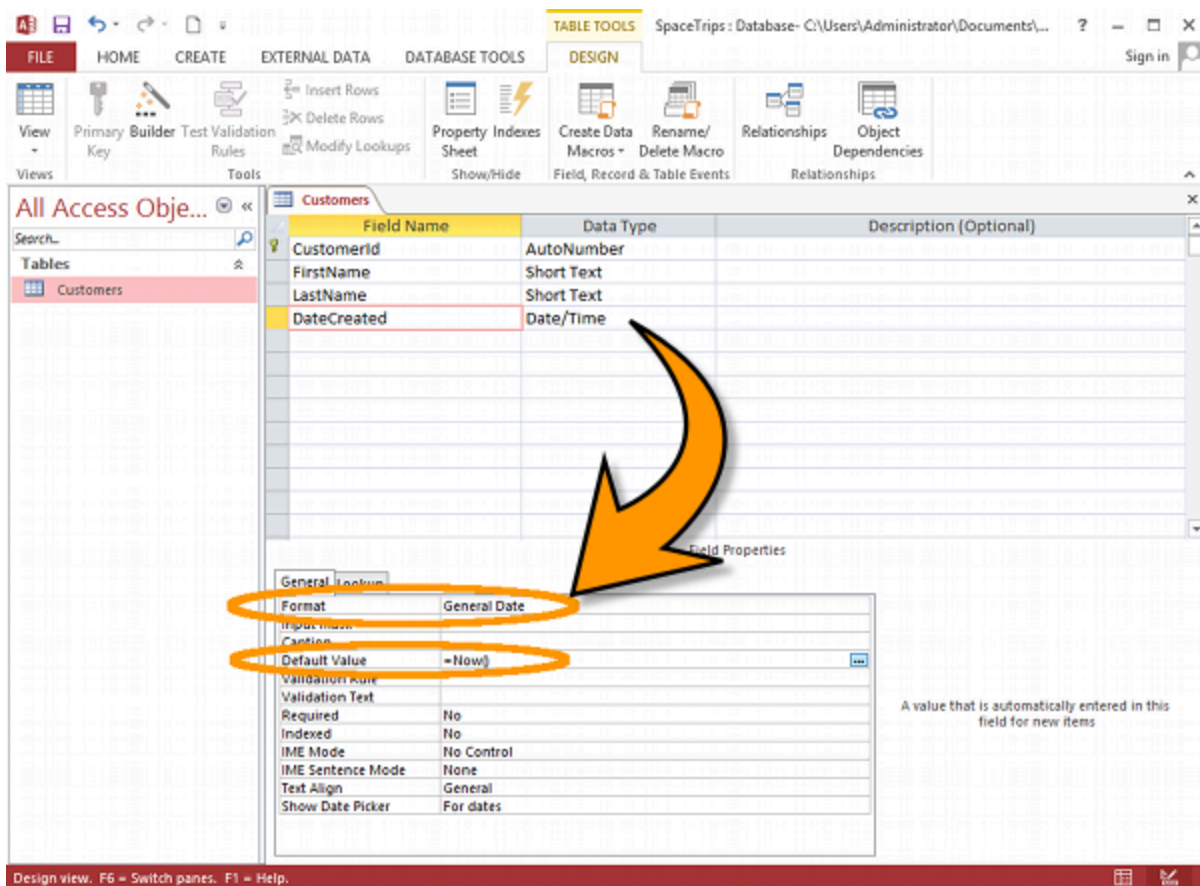
Field Name	Data Type	Description (Optional)
AlbumId	AutoNumber	
AlbumName	Short Text	
ReleaseDate	Date/Time	
ArtistId	Number	
GenreId	Number	

Field Properties

Property	Value
Field Size	255
Format	
Input Mask	
Caption	
Default Value	
Validation Rule	
Validation Text	
Required	Yes
Allow Zero Length	No
Indexed	No
Unicode Compression	Yes
IME Mode	No Control
IME Sentence Mode	None
Text Align	General

Require data entry in this field?

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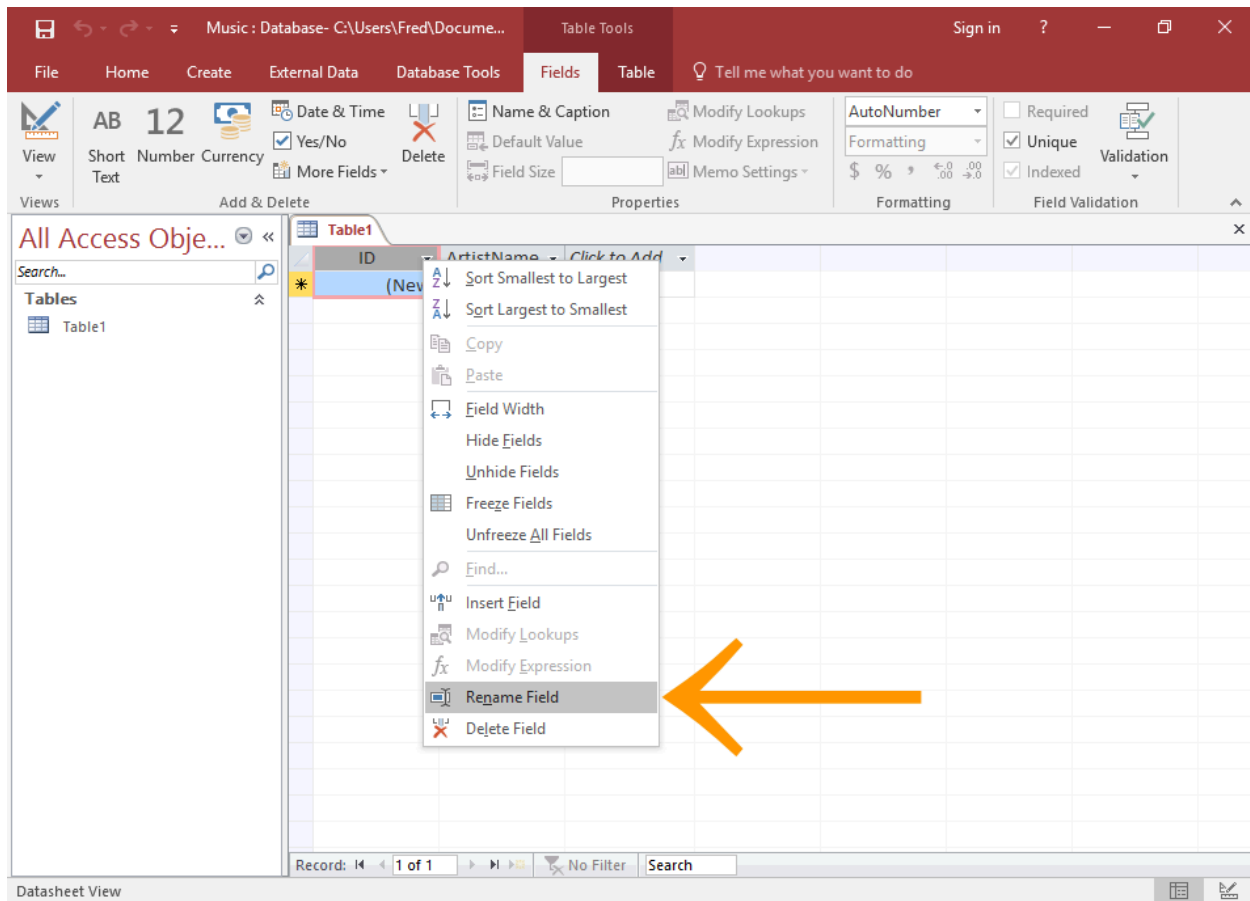


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Entering Data in a Table, Preview and Print a Table, Changing Row/Column Height

- Switch to **Datasheet View** → Type data directly (like Excel).
- **Change Row/Column Height** → Drag borders between rows/columns.
- **Preview & Print** → File → Print → Print Preview shows table layout.

Here are examples of entering data in **Datasheet View**:



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How to Create a Table in Datasheet View in Access 2016

Closing and Opening of Table, Sorting of Table, Finding and Replacing Texts

- **Close Table** → Click × on table tab.
- **Open Table** → Double-click table name in Navigation Pane (left side).
- **Sorting** → Click column header → Right-click → Sort A to Z / Z to A.
- **Find/Replace** → Home tab → Find (or Ctrl + F) → Find/Replace tab.

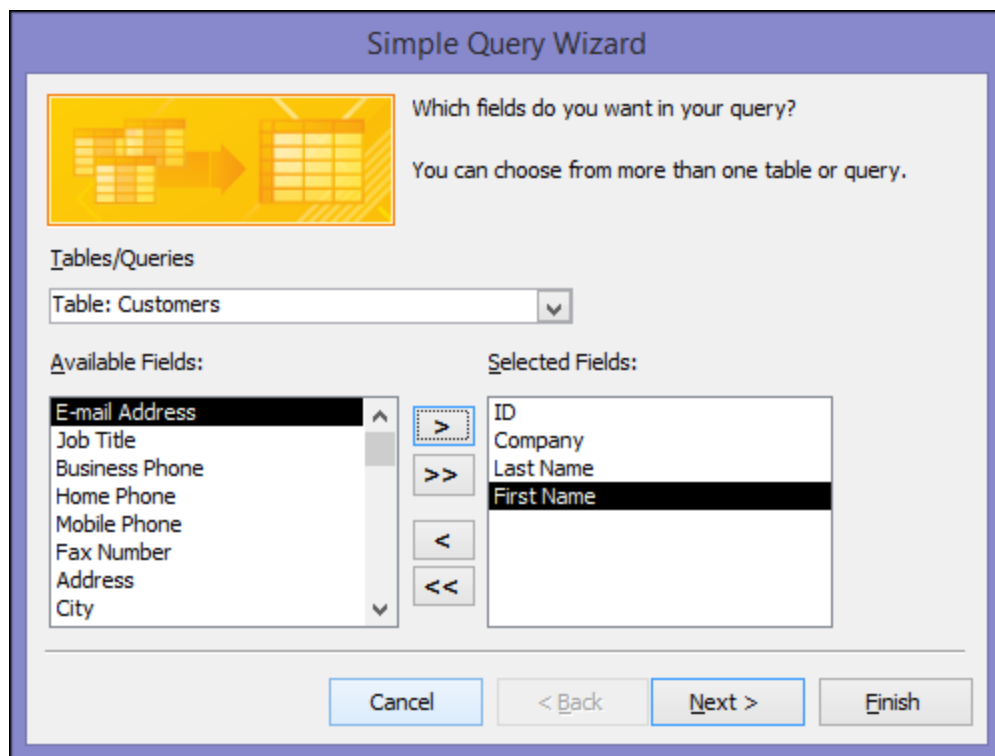
Using Queries Wizard

Queries help filter, sort, and show specific data.

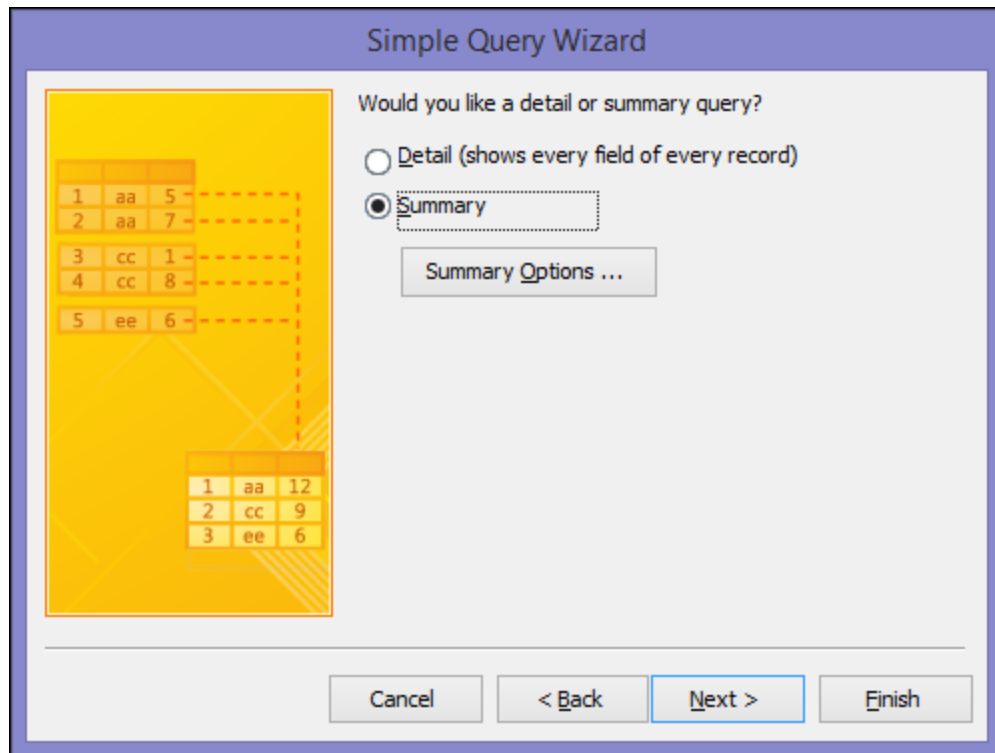
1. Create tab → Query Wizard.

2. Choose **Simple Query Wizard**.
3. Select table → Add fields → Next → Finish.
4. View results in Datasheet View.

Here are step-by-step examples of the **Query Wizard** in action:



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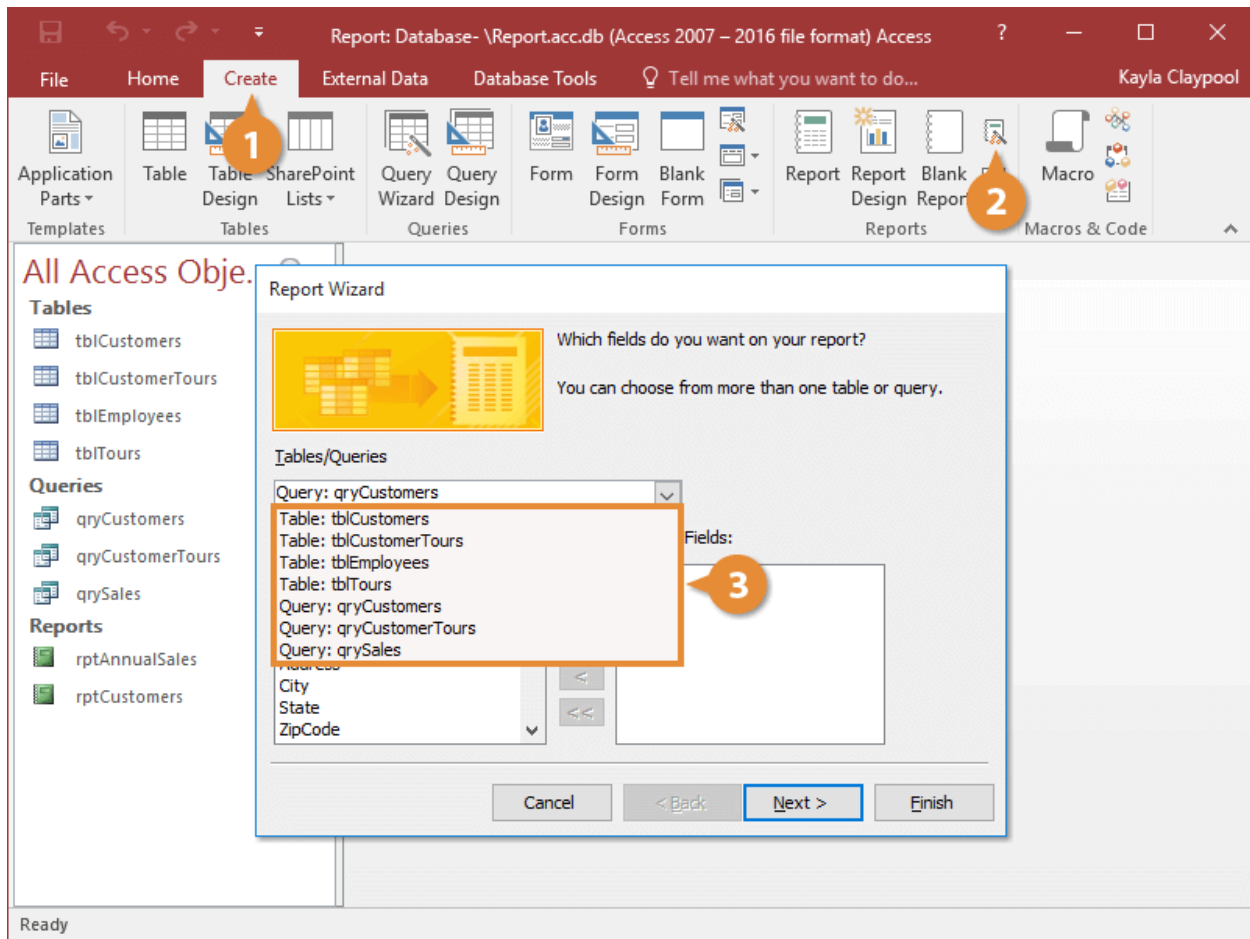
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Creating Report from Tables/Queries from Report Wizard, Modifying a Report, Printing of Report

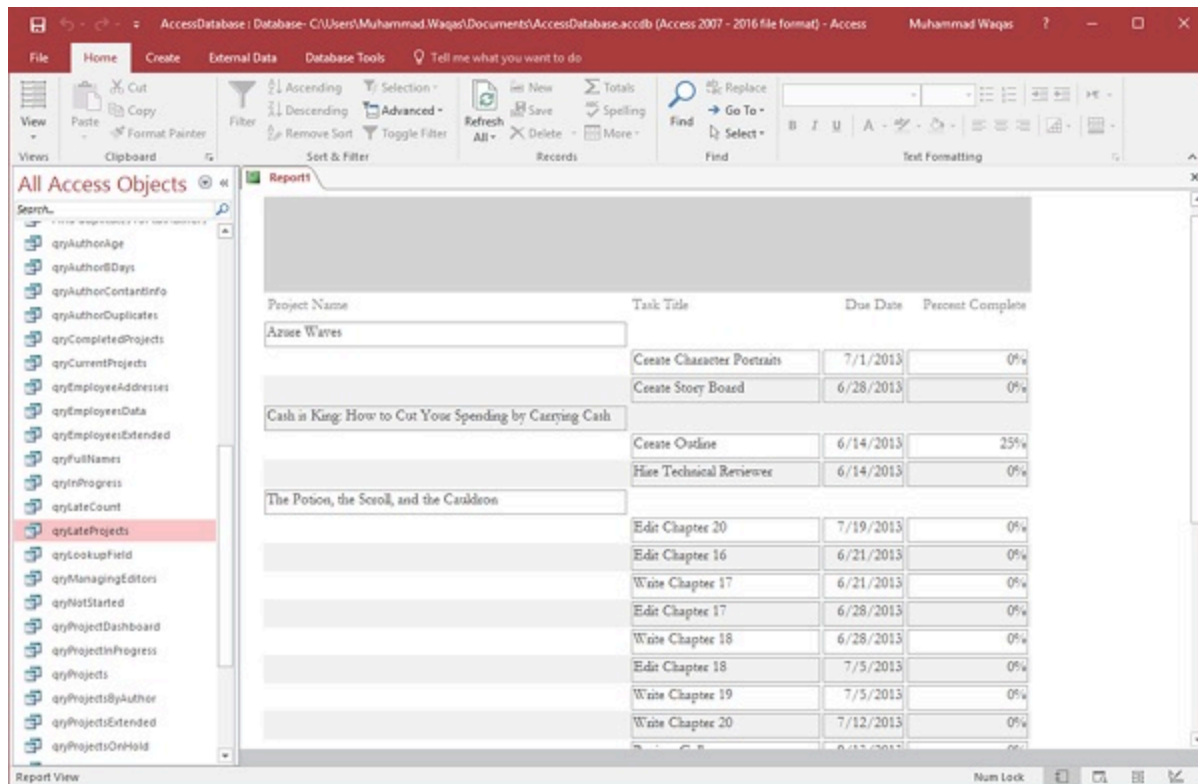
Reports present data nicely for printing.

1. Create tab → Report Wizard.
2. Select table/query → Add fields → Group/Sort options → Layout → Finish.
3. **Modify** → In Layout/Design View → Add labels, change fonts/colors.
4. **Print** → File → Print.

Here are examples of the **Report Wizard** steps and a finished formatted report:



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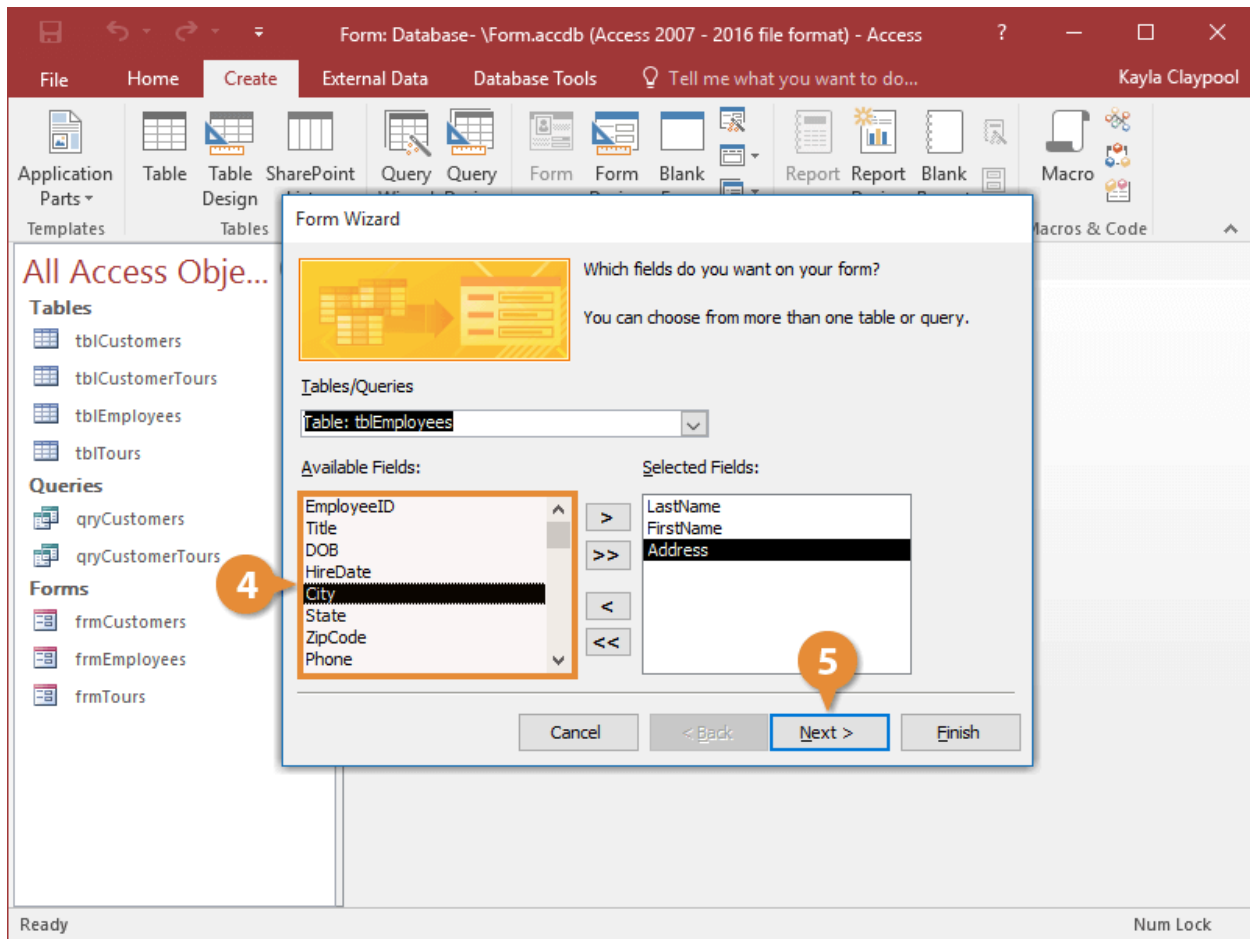
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Creating a Form Using Wizard, Entry in the Forms, Basics of Formatting of Forms and Reports

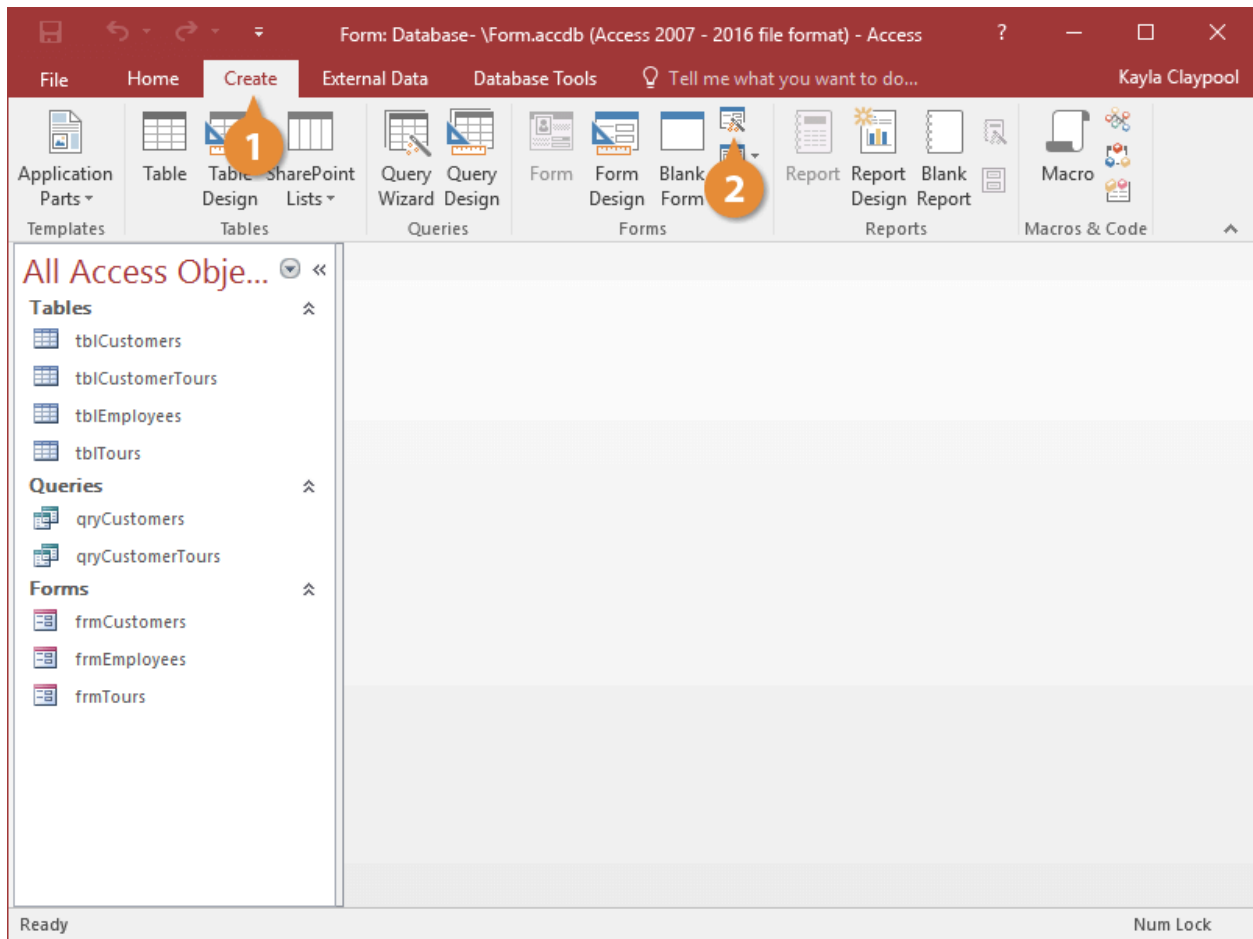
Forms make data entry easy and attractive.

1. Create tab → Form Wizard.
2. Select table/query → Add fields → Layout (Columnar, Tabular, etc.) → Finish.
3. **Entry** → Open form → Use navigation buttons (New Record, Next, Previous).
4. **Formatting** → In Layout/Design View → Change colors, fonts, add logos, adjust sizes.

Here are visual examples of the **Form Wizard** and a created form:



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